Suggested Reading: Linear Algebra Foundations

Vectors, Norms, Matrices, Rank, Orthogonality, and Projection

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Suggested Reading

For the topics covered so far — vectors, norms, matrices, linear independence, basis and rank, orthogonality, and projection — the primary reference is *Mathematics for Machine Learning* (MML). Relevant sections are §§2.1–2.2 and §2.4 for vectors, matrices, and vector spaces; §§3.1 and 3.3 for norms, lengths, and distances; §§2.5–2.7 for linear independence, basis, rank, and linear mappings; and §§3.2, 3.4, and 3.8 for inner products, orthogonality, and projections. Section 3.5, on orthonormal bases, may be treated as optional enrichment.

For a more concrete and computational treatment, consult *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares* (VMLS), particularly Chapters 1, 3, 5, and 6. Within Chapter 5, Section 5.3 (orthonormal vectors) and Section 5.4 (the Gram–Schmidt algorithm) may both be treated as optional enrichment, alongside MML §3.5 above — this is the same material covered from two different angles.

You are not expected to read every listed section completely or to adopt notation that differs from the lectures. Use the lecture slides as the guide to the required scope, and use these books to reinforce concepts, study additional examples, and resolve points that remain unclear. Try to understand the material as one connected story: vectors represent objects; norms measure their size or difference; matrices combine and transform vectors; rank measures independent information; orthogonality separates directions; and projection finds the closest approximation within a subspace. Working through a few exercises is generally more valuable than reading many pages passively.

Quick reference, by topic

Topic	MML (primary)	VMLS (secondary)
Vectors, vector spaces	§2.1–2.2, §2.4	Ch. 1
Norms, length, distance	§3.1, §3.3	Ch. 3

Topic	MML (primary)	VMLS (secondary)
Matrices, linear independence, basis, rank, linear mappings	§2.2, §2.5–2.7	Ch. 5, Ch. 6
Orthogonality, projection	§3.2, §3.4, §3.8	—
Optional: orthonormal bases, Gram–Schmidt	§3.5	§5.3–5.4